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$$f(x) = \tan(2 \operatorname{Bgcot} x)$$

neem $y = \operatorname{Bgcot} x$ dan is $\cot y = x$ met $y \in]0, \pi[$

$$\cot y = \frac{\cos y}{\sin y} = x \text{ dan is } \tan y = \frac{\sin y}{\cos y} = \frac{1}{\cot y} = \frac{1}{x}$$

$$\text{we gebruiken de identiteit } \tan 2y = \frac{2 \tan y}{1 - \tan^2 y}$$

$\tan 2y$ is enkel gedefinieerd als $2y \neq \frac{\pi}{2} + k\pi$ met $k \in \mathbb{Z}$

$$\Rightarrow 2y \neq \frac{\pi}{2} \text{ en } 2y \neq \frac{3\pi}{2} \text{ want } 2y \in]0, 2\pi[$$

$$\Rightarrow y \neq \frac{\pi}{4} \text{ en } y \neq \frac{3\pi}{4} \text{ want } y \in]0, \pi[$$

$$\Rightarrow x \neq \cot \frac{\pi}{4} \text{ en } x \neq \cot \frac{3\pi}{4} \text{ oftewel } x \neq \pm 1$$

$$\tan 2y = \frac{2 \tan y}{1 - \tan^2 y}$$

$$= \frac{2 \cdot \frac{1}{x}}{1 - \left(\frac{1}{x}\right)^2}$$

$$= \frac{\frac{2}{x}}{1 - \frac{1}{x^2}}$$

$$= \frac{\frac{2}{x}}{\frac{x^2 - 1}{x^2}}$$

$$= \frac{2}{x} \cdot \frac{x^2}{x^2 - 1}$$

$$= \frac{2x^2}{x(x^2 - 1)}$$

$$= \frac{2x}{x^2 - 1}$$

$$\text{dus is } \tan(2 \operatorname{Bgcot} x) = \frac{2x}{x^2 - 1} \text{ met } x \neq \pm 1$$

References